



Biography

I am a Full Professor at the Department of Electronics, Information and Bioengineering of Politecnico di Milano and a member of its Artificial Intelligence and Robotics Laboratory.

My research lies at the intersection of robotics, machine learning, computer vision, and pattern recognition. I develop and evaluate learning and perception methods for autonomous physical and software systems, especially when they must act under uncertainty or with incomplete knowledge.

Autonomous robots, intelligent vehicles, agricultural robotics, assistive technologies, and intelligent data analysis provide the real-world settings for this work. I am also deeply involved in teaching, doctoral supervision, research coordination, benchmarking, and technology transfer.

Research Interests

- **Robotics and Autonomous Systems** — Perception, navigation, mapping, sensor fusion, planning, and learning for autonomous robots and intelligent vehicles.
- **Machine Learning and Pattern Recognition** — Adaptive models, representation learning, probabilistic methods, and reliable experimental methodologies for intelligent systems.
- **Computer Vision** — Visual perception, event-based vision, 3D reconstruction, scene understanding, and multimodal learning.
- **AI for the Physical World** — Applied AI for agriculture, assistive technology, smart eyewear, transportation, medicine, and industrial systems.

Publications

12490 citations · h-index 52 · [Google Scholar](#)

Journal Articles

- [1] Uselli, Mirko, Rapado-Rincon, David, Kootstra, Gert, Matteucci, Matteo (2026). AgriGS-SLAM: Orchard Mapping Across Seasons via Multi-View Gaussian Splatting SLAM. IEEE Robotics and Automation Letters.
- [2] Plizzari, Chiara, Cannici, Marco, Matteucci, Matteo (2021). Skeleton-Based Action Recognition via Spatial and Temporal Transformer Networks. Computer Vision and Image Understanding, 208, pp. 103219. [10.1016/j.cviu.2021.103219](https://doi.org/10.1016/j.cviu.2021.103219)
- [3] Bertoglio, Riccardo, Corbo, Chiara, Renga, Filippo M., Matteucci, Matteo (2021). The Digital Agricultural Revolution: A Bibliometric Analysis Literature Review. IEEE Access, 9, pp. 134762–134782. [10.1109/ACCESS.2021.3115258](https://doi.org/10.1109/ACCESS.2021.3115258)
- [4] Vali, Ava, Comai, Sara, Matteucci, Matteo (2020). Deep Learning for Land Use and Land Cover Classification Based on Hyperspectral and Multispectral Earth Observation Data: A Review. Remote Sensing, 12, pp. 2495. [10.3390/rs12152495](https://doi.org/10.3390/rs12152495)
- [5] Bonarini, Andrea, Matteucci, Matteo, Migliavacca, Martino, Rizzi, Davide (2014). R2P: An Open Source Hardware and Software Modular Approach to Robot Prototyping. Robotics and Autonomous Systems, 62, pp. 1073–1084. [10.1016/j.robot.2013.08.009](https://doi.org/10.1016/j.robot.2013.08.009)
- [6] Maggi, Federico, Matteucci, Matteo, Zanero, Stefano (2010). Detecting Intrusions through System Call Sequence and Argument Analysis. IEEE Transactions on Dependable and Secure Computing, 7, pp. 381–395. [10.1109/TDSC.2008.69](https://doi.org/10.1109/TDSC.2008.69)

- [7] Ceriani, Simone, Fontana, Giulio, Giusti, Alessandro, Marzorati, Daniele, Matteucci, Matteo, Migliore, Davide, Rizzi, Davide, Sorrenti, Domenico G., Taddei, Paolo (2009). RAWSEEDS Ground Truth Collection Systems for Indoor Self-Localization and Mapping. *Autonomous Robots*, 27, pp. 353–371. [10.1007/s10514-009-9156-5](https://doi.org/10.1007/s10514-009-9156-5)

Conference Papers

- [1] Sbrolli, Cristian, Matteucci, Matteo (2026). SCENEFORGE: Enhancing 3D-Text Alignment with Structured Scene Compositions. *Advances in Neural Information Processing Systems*, 38, pp. 84731–84750.
- [2] Long, Juncen, Bardaro, Gianluca, Mentasti, Simone, Matteucci, Matteo (2026). DGCMam: Fusing Distance Graph Convolution and Mamba for Skeleton-Based Action Recognition. *IEEE International Conference on Automatic Face and Gesture Recognition*.
- [3] Cannici, Marco, Ciccone, Marco, Romanoni, Andrea, Matteucci, Matteo (2020). A Differentiable Recurrent Surface for Asynchronous Event-Based Data. *European Conference on Computer Vision*, pp. 136–152. [10.1007/978-3-030-58598-3_9](https://doi.org/10.1007/978-3-030-58598-3_9)
- [4] Cannici, Marco, Ciccone, Marco, Romanoni, Andrea, Matteucci, Matteo (2019). Asynchronous Convolutional Networks for Object Detection in Neuromorphic Cameras. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*.
- [5] Visin, Francesco, Ciccone, Marco, Romero, Adriana, Kastner, Kyle, Cho, Kyunghyun, Bengio, Yoshua, Matteucci, Matteo, Courville, Aaron (2016). ReSeg: A Recurrent Neural Network-Based Model for Semantic Segmentation. *IEEE Conference on Computer Vision and Pattern Recognition Workshops*.

Experience

2021–present	Full Professor Department of Electronics, Information and Bioengineering, Politecnico di Milano · Milan, Italy RESEARCH — Robotics, machine learning, computer vision, pattern recognition, and intelligent systems.
2015–2021	Associate Professor Department of Electronics, Information and Bioengineering, Politecnico di Milano · Milan, Italy
2005–2015	Assistant Professor Department of Electronics and Information, Politecnico di Milano · Milan, Italy
2003–2005	Research Assistant Department of Electronics and Information, Politecnico di Milano · Milan, Italy
2001–2002	Visiting Student Center for Automatic Learning and Discovery, Carnegie Mellon University · Pittsburgh, PA, USA

Projects

2023–2027	AgrifoodTEF — Testing and Experimentation Facilities for the Agri-Food Domain European Commission · Digital Europe · European project · Politecnico di Milano ROLE — Principal Investigator for Politecnico di Milano A European network of real-world facilities for testing and validating AI and robotics solutions in the agri-food domain.
2021–2023	AI-SPRINT — Artificial Intelligence in Secure PRivacy-preserving computing coNTinuum European Commission · Horizon 2020 · European project · Politecnico di Milano ROLE — Co-Principal Investigator for the coordinating institution Tools for secure, privacy-preserving AI applications across the edge-to-cloud computing continuum.

2020–2022	METRICS — Metrological Evaluation and Testing of Robots in International Competitions European Commission · Horizon 2020 · European project · Politecnico di Milano ROLE — Principal Investigator for Politecnico di Milano Reproducible evaluation and benchmarking of robots, including agri-food competitions.
2019–2022	DeepField — Deep Learning in Field Robotics European Commission · Horizon 2020 · European project · Politecnico di Milano ROLE — Principal Investigator for Politecnico di Milano Deep-learning methods for precision agriculture and field robotics.
2018–2021	SciRoc — European Robotics League plus Smart Cities Robot Competitions European Commission · Horizon 2020 · European project · Politecnico di Milano ROLE — Principal Investigator for Politecnico di Milano
2011–2013	ROAMFREE — Robust Odometry Applying Multisensor Fusion to Reduce Estimation Errors Italian Ministry of University and Research · PRIN 2009 · National project · Politecnico di Milano ROLE — National Scientific Coordinator Robust robot odometry through multisensor fusion.
2006–2009	RAWSEEDS — Robotics Advancement through Web-publishing of Sensorial and Elaborated Extensive Data Sets European Commission · FP6 · European project · Politecnico di Milano ROLE — Project Coordinator A benchmarking toolkit with multisensor datasets, validated ground truth, and evaluation methods for mobile-robot localization and mapping.

Education

2000–2003	Ph.D. in Computer Engineering and Automation Politecnico di Milano · Milan, Italy THESIS — Evolutionary Learning of Adaptive Models within a Bayesian Framework
2001–2002	Master of Science in Knowledge Discovery and Data Mining Carnegie Mellon University · Pittsburgh, PA, USA THESIS — ELearNT: Evolutionary Learning of Rich Neural Network Topologies
1993–1999	Laurea in Computer Engineering Politecnico di Milano · Milan, Italy THESIS — Rappresentazione della conoscenza fuzzy e a intervalli per algoritmi di apprendimento per rinforzo applicati ad agenti situati

Skills

- **Research** — Robotics, Machine Learning, Computer Vision, Pattern Recognition, Signal Processing
- **Autonomous Systems** — Perception, SLAM, Sensor Fusion, Robot Learning, Benchmarking
- **Application Domains** — Intelligent Vehicles, Agricultural Robotics, Assistive Technologies, Event-based Vision, Intelligent Data Analysis
- **Languages** — Italian (native), English (fluent)

Teaching Activities

2026	Machine Learning Lecturer and Coordinator · Politecnico di Milano · MSc in Bioinformatics for Computational Genomics · Coordinator since 2020
2026	Artificial Neural Networks and Deep Learning Lecturer and Coordinator · Politecnico di Milano · MSc in Computer Science and Engineering · Coordinator since 2019

2026	Robotics Lecturer and Coordinator · Politecnico di Milano · MSc and BSc in Computer Science and Engineering · Coordinator since 2016
2020	Machine Learning for Non-Matrix Data Lecturer · Politecnico di Milano · Ph.D. in Information Technology
2019	Advances in Deep Learning with Applications in Text and Image Processing Lecturer and Coordinator · Politecnico di Milano · Ph.D. in Information Engineering
2018	Deep Learning: Theory, Techniques and Applications Lecturer and Coordinator · Politecnico di Milano · Ph.D. in Information Engineering

Supervision

2026	Cristian Sbrolli Ph.D. student · Advisor Multimodal and generative learning for visual understanding
2026	Nico Catalano Ph.D. student · Advisor Few-shot learning and computer vision for precision agriculture
2026	Eugenio Lomurno Ph.D. student · Advisor Reliable machine learning, synthetic data, and privacy
2026	Matteo Frosi Ph.D. student · Advisor 3D perception and SLAM with learned priors
2022	Marco Cannici Ph.D. · Advisor Learning Efficient and Effective Representations for Event-based Cameras
2022	Simone Mentasti Ph.D. · Advisor Embedded AI and sensor fusion for autonomous vehicles
2017	Andrea Romanoni Ph.D. · Advisor Incremental Large-Scale Visual 3D Mesh Reconstruction
2017	Francesco Visin Ph.D. · Advisor Deep recurrent neural networks for visual scene understanding
2014	Martino Migliavacca Ph.D. · Co-advisor The R2P framework for robot prototyping
2011	Luigi Malagò Ph.D. · Advisor On the geometry of optimization based on the exponential family relaxation

Talks

2024	La percezione dei robot ai tempi dell'AI: precisione o flessibilità? BI-MU and I-RIM · Milan, Italy · Invited talk
2018	Autonomous Driving and Artificial Intelligence AlxIA 2018 · Trento, Italy · Invited talk

- 2017 **Do Big Data Make Common Sense?**
Big Data e Robotica nella fabbrica digitale connessa · Milan, Italy · Invited talk
- 2016 **Modular Development of Service Robots**
A&T — Robot & Co. · Turin, Italy · Invited talk
- 2016 **Robotics Benchmarking and Good Experimental Methodologies**
European Robotics Forum · Ljubljana, Slovenia · Invited talk

Awards and Achievements

- 2025 **NVIDIA Academic Grant Award**
NVIDIA · Four RTX PRO 6000 Blackwell Max-Q Workstation Edition GPUs
- 2025 **Leonardo Fibonacci Best Paper Award**
11th International Conference on Machine Learning, Optimization, and Data Science · Neuro-Symbolic Scene Graph Conditioning for Synthetic Image Dataset Generation
- 2020 **Outstanding Reviewer Award**
British Machine Vision Conference
- 2019 **Best Paper Award**
CVPR Workshop on Event-based Vision and Smart Cameras · Asynchronous Convolutional Networks for Object Detection in Neuromorphic Cameras
- 2016 **Best Paper Award**
DeepVision at CVPR · ReSeg: A Recurrent Neural Network-based Model for Semantic Segmentation
- 2011 **Antonio D'Auria Prize and Intesa Sanpaolo Startup Initiative**
RobyWheelChair project
- 2003 **Dimitris N. Chorafas Foundation Award**
Best Ph.D. thesis
- 2001 **Rotary Foundation Ambassadorial Scholarship**
Rotary Foundation

Data Authorization

In compliance with Italian Legislative Decree no. 196 of 30 June 2003 and the GDPR (EU 2016/679), I authorize the use and processing of the personal data contained in this curriculum vitae.